1. Native chained Jacobian transfer over F_{101}

2026-11-11 · continuation of crypto PR 3877 · queue issue 3904

Standing: unverified producer assertions. Local correctness checks and kernel proof builds are retained. There is no independent signed validation and no solving-cost benchmark. Full report and sources: REPORT_20261111.md and SOURCES_20261111.md beside this PDF.

1.1. Explicit route

The source is E: $v^2=u^3+1$ with P=(4,41), order 17. The degree-two cover f: C → E is $f(x,y)=(x^2,y)$, C: $y^2=x^6+1$. Its whole-fiber pullback enters Jac(C). The ordered quadratic splitting $(x^2+1)(x^2+10x-1)(x^2-10x-1)$ has determinant 40 in F_{101}. Kunzweiler's Richelot correspondence gives a degree-four (2,2)-isogeny to $Jac(C_2)$, C_2: $w^2=41z^6+9z^4+9z^2+41$.

The first quotient of C_2, with $X=41z^2$ and $V=41w$, reaches E_2: $V^2=X^3+9X^2+66X+39$. The local test obtains (91,84) from P. All 17 source scalars and all 289 subgroup addition pairs are checked. The second quotient sends this image subgroup to zero; it remains an explicit negative control. [pcc:unverified]

1.2. Native maps and exceptional points

The source uses the odd-degree chart $t=1/(x-10)$, $Y=yt^3$: $F(t)=1+60t+86t^2+2t^3+15t^4+60t^5$. Canonical Mumford pairs represent degree-zero classes based at W=(10,0). This avoids using one-infinity Cantor arithmetic directly on a sextic.

Whole-fiber pullback handles nonsplit and ramified fibers and the degree drop at (−1,0). Both source quotient norms accept every reduced rational class, split quadratic support in $F_{101}[s]/(s^2-2)$, retain multiplicity, and handle the two original infinities. Elliptic infinity and unsupported input are different API outcomes. [pcc:unverified]

The Richelot evaluator computes polynomial fibers over F_{101^2}, composes them on an odd target chart, and subtracts the source-base-point image. The extended version admits all 10,404 rational source classes, including Weierstrass, source infinity and degree-drop cases. Kernel size is four; the 2,601 image classes each have four preimages in the local checks. This is fixed-field coverage; map evaluation never calls a DLP solver. [pcc:unverified]

1.3. Development evidence

Check	Scope and evidence
Native whole-fiber maps	102 elliptic inputs; all ordered additivity pairs
Source classes	10,404 canonical classes; both norms and three translations
Differential reference	10,404 additions/inverses, unchanged cryptanalysis BigUint code
Richelot chain	17 scalars; 289 addition pairs; nonzero and zero endpoint controls
Dual norm	All 10,404 classes: dual-after-forward equals multiplication by two
Elliptic shortcut	All 102 inputs: full route equals τ φ after doubling
Lean structure	8 structural + 6 polynomial lemmas; 20+6 declarations audited

These are local development checks, not independent signed validation. Both Jacobian implementations use Cantor arithmetic. Raw receipts and failed attempts live in evidence/; claim standings remain unverified.

1.4. Proof map and prime-field boundary

Core Lean 4.35.0 is pinned. The two-hop norm lemma multiplies the individual dual norm identities. A Bézout certificate and the actual source subgroup annihilator then give recovery and injectivity. Anti-invariance yields two-torsion; odd subgroup exponent yields zero. The existence of the geometric objects and their norm identities is explicit input, not proved by these structural lemmas. Six further polynomial lemmas prove the cleared source and target identities, the F101 chart, and the elliptic endpoint isomorphism/inverse. Only the standard foundations propext, Classical.choice and Quot.sound occur. [pcc:unverified]

A separate lemma shows that a unital additive map from characteristic p to characteristic q requires q dividing p. This excludes field maps for distinct prime characteristics. It says nothing about arbitrary efficient maps between encoded cyclic subgroups. A prime field has no proper subfield, so ordinary Weil restriction does not provide a smaller-base-field route. Lifting and cross-prime reduction remain separate open mechanisms. [pcc:unverified]

Claims: PCC-20261111-ca60cb (native maps), PCC-20261111-3f548f (Richelot fixture), PCC-20261111-7a3fe0 (conditional chain injectivity), PCC-20261111-a00fb9 (characteristic boundary). All are unverified until a signed verdict establishes otherwise. Geometry, exact Richelot kernel, and the full Picard bridge are not yet formalized in this file.

1.5. Reproduction and unresolved work

From crypto, add Cargo and Lean to PATH and run the native Cargo tests, then tools/lean/proof-gate.sh on lean/ChainTransfer.lean. The root examples chained_jacobian_reference and chained_jacobian_identities provide the BigUint comparison and denominator-cleared identity certificates. The cross-repository oracle is cryptanalysis:experiments/chained-transfer-reference.

Retain Lean attempt 01 (failed elaboration) and root-library attempts 01–03 (off-disk dependencies); never treat them as passes. The canonical queue and resume state record the latest build outcomes and exact next actions.

No remote experiment has been launched: S3 checkpoint transport is unavailable and the live remaining IsoLab budget/capacity is unknown. The research-KB MCP/CLI is unavailable, and crypto-autoresearcher is vault locked without its key binding. Local unit tests and proof development remain possible.

The full chain equals a cheaper expression on all 102 source elliptic points: $\tau\ \varphi$ after doubling, where φ is the historical 2-isogeny and $\tau(U,V)=(82U+60,98V)$. The reverse correspondence, with its sign corrected, provides multiplication by two on every rational source Jacobian class. Source norm then gives multiplication by four, invertible modulo 17. Point recovery is distinct from solving its unknown scalar. [pcc:unverified]

Next obligations: publish the full target-side differential evidence, obtain an independent signed validator, generalize the fixed-field backend, and freeze a total-cost comparison. That comparison must include the collapsed elliptic route as a separate arm. There is no measured speedup in this report.

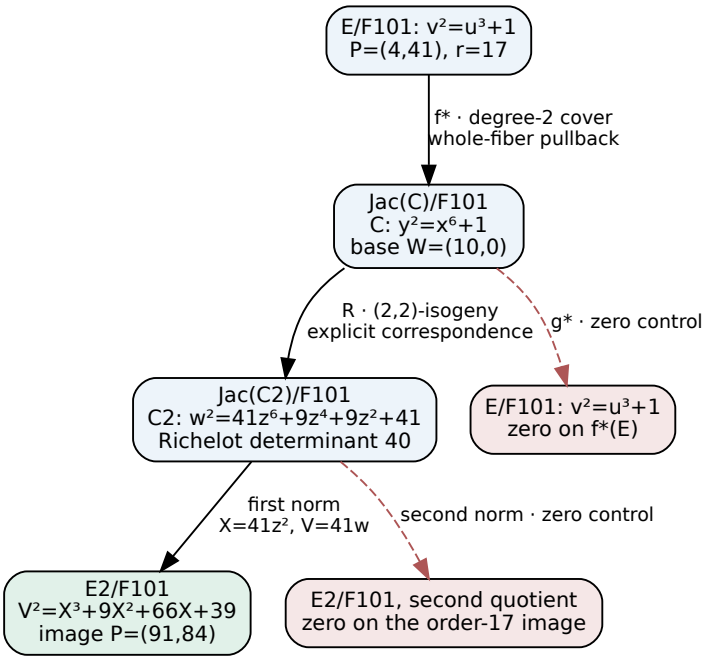
1.6. Primary sources

- S. Galbraith, Mathematics of Public Key Cryptography, corrected version 2.0, ch. 8 §8.3 and ch. 10 §§10.3–10.4, author-hosted full PDFs.
- S. Kunzweiler, Efficient computation of $(2^n,2^n)$-isogenies, Designs, Codes and Cryptography 92 (2024), 1761–1802. DOI 10.1007/s10623-024-01366-1, §3.1–3.2, Propositions 7–8 and Algorithm 1. Full publisher PDF read.

The local route diagram has no leaderboard promotion. Canonical transfer and IC graph sources remain unchanged pending their evidence gates.

2. Explicit route and zero controls

F101 chained Jacobian fixture · producer development evidence · PCC unverified



All nodes and maps are over F_{101}. The solid path is the explicit development fixture; dashed edges are the retained vanishing projections. No edge denotes a measured computational advantage. [pcc:unverified]